INDIAN INSTITUTE OF TECHNOLOGY KANPUR
DEPARTMENT OF SPASE
SPA617M: Space Instrumentation Lab

Experiment Title & No.:	Exp 2: RF Components: Characterization and Measurement
Course Code & Title:	SPA617M - Space Instrumentation Lab
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Lab Dates:	Date of Experiment: August 19 + Sept. 2, 2026

Note: I have taken help from AI to make graphs and structure and all, but my understanding section is genuine

1. AIM

To understand the fundamental operational principles and experimentally characterize key radio-frequency (RF) components—including an RF filter, mixer, power splitter, power splitter + combiner and amplifier used in an RF receiver chain.

2. OBJECTIVES

- To characterize an RF filter, measure its transmission response (insertion loss vs. frequency), identify its filter type, and determine its cutoff frequencies / passband.
- To estimate the input voltage reflection coefficient (|Γ|) and return loss (RL) of the RF filter using a Tee-connector setup on an oscilloscope.
- To demonstrate the filtering behavior of a quarter-wave (λ/4) open-circuit coaxial stub acting as a notch filter and compare theoretical vs. measured resonance.
- To observe heterodyne frequency conversion in an RF mixer and quantify its conversion loss across sum and difference intermediate frequencies (IF).
- To measure the power division ratio, insertion loss, and isolation symmetry of a 2-way RF power splitter and power combiner and of splitter and combiner together.
- To measure the power loss in splitter + combiner configuration and to measure total loss vs frequency of generator for the same configuration.
- To characterize an RF amplifier: measure small-signal gain vs. frequency, observe harmonic generation, and determine the 1-dB compression point (P1dB).
- To evaluate non-linear intermodulation distortion via a two-tone amplifier test and determine the Third-Order Intercept (TOI / IIP3 / OIP3) point.

3. THEORETICAL BACKGROUND & MATHEMATICAL FORMULATIONS

3.1 RF components in an RF receiver chain

A minimal RF receiver chain is shown in Fig. 1. A filter selects or rejects specified frequency ranges. A mixer translates a signal from one frequency to another. A low-noise amplifier (LNA) amplifies weak received signals while adding as little noise as possible. A power splitter/combiner distributes a signal to multiple paths or combines signals from different paths.

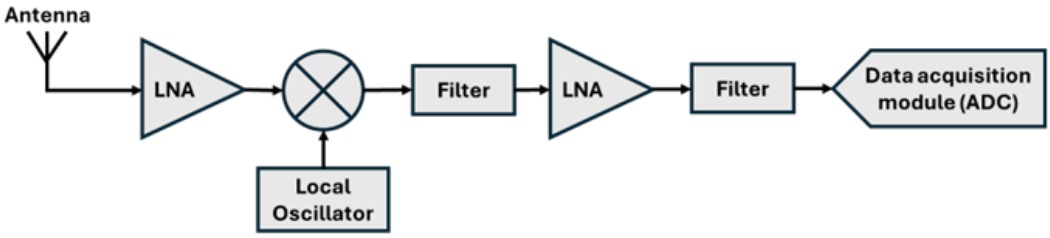


Figure 1: Minimal RF receiver front-end chain in radio astronomy

3.2 Filter transmission characteristics

A filter passes selected frequency components and attenuates others. The four commonly used filter types are:

- **Low-pass filter:** passes frequencies below its cutoff frequency and attenuates higher frequencies.
- **High-pass filter:** passes frequencies above its cutoff frequency and attenuates lower frequencies.
- **Band-pass filter:** passes frequencies between its lower and upper cutoff frequencies.
- **Band-reject or notch filter:** attenuates frequencies within a specified stopband and passes frequencies outside it.

For low-pass and high-pass filters, fc denotes the cutoff frequency and fs denotes the stopband edge. For band-pass and notch filters, f1 and f2 denote the lower and upper cutoff frequencies, while f0 denotes the center frequency or notch frequency.

For the transmission measurement, let Pin be the power measured without the filter and Pout be the power measured after inserting the filter. The insertion loss is

$$IL = Pin - Pout \ dB$$

A smaller value of IL indicates better transmission through the filter. The 3 dB cutoff frequencies are measured relative to the minimum insertion loss in the passband.

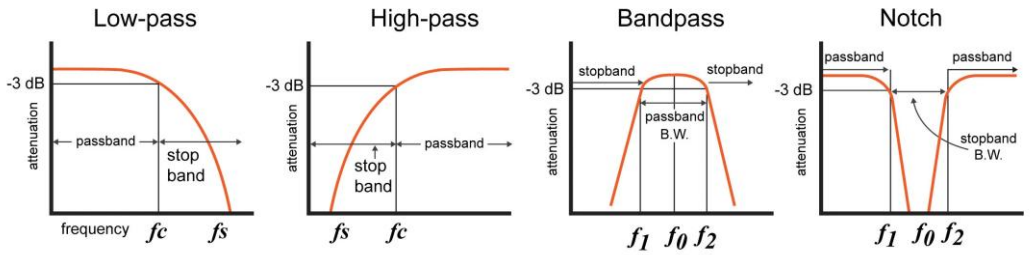


Figure 2: Transmission responses of common RF filter types (Low-pass, High-pass, Band pass, and Notch).

3.3 Filter reflection and return loss

When the input impedance of the filter, Zin, is not equal to the characteristic impedance of the RF system, Z0, a part of the incident signal is reflected from the filter input. The voltage reflection coefficient is defined as

$$\Gamma = Vref / Vinc = (Zin - Z0) / (Zin + Z0)$$

Where Vinc and Vref are the incident and reflected voltage amplitudes, respectively. The magnitude indicates the fraction of the incident voltage amplitude that is reflected. For a perfectly matched filter, Zin = Z0 and $|\Gamma| = 0$. For complete reflection, $|\Gamma| = 1$.

The return loss is

$$RL = -20 \log_{10} |\Gamma| \ dB$$

A well-matched filter has a small reflection coefficient and a large return loss. A poorly matched filter has a larger reflection coefficient and a smaller return loss. The important limiting cases are:

Condition	$\|\Gamma\|$	RL
Perfect match	0	∞ dB
Partial mismatch	$0 < \|\Gamma\| < 1$	> 0 dB
Complete reflection	1	0 dB

3.4 Quarter-wave open-circuit stub filter

An open-ended coaxial stub connected in shunt through a Tee connector behaves as a resonator. At its quarter-wave resonant frequency, the open circuit at the far end is transformed approximately into a low impedance at the Tee junction. The signal is therefore strongly attenuated, producing a notch.

The physical length of the stub is approximately

$$L = \lambda / 4 = (VF \cdot c) / (4 f0)$$

Where L is the stub length, λ is the guided wavelength in the cable, VF is the velocity factor, c is the speed of light in vacuum, and f0 is the expected notch frequency. The connector length and open-end effect may cause the measured value of f0 to differ slightly from the calculated value.

3.5 Mixer

An RF mixer is a three-port nonlinear device with RF, local-oscillator (LO), and intermediate-frequency (IF) ports. When signals at fRF and fLO are applied, the mixer produces the sum and difference components:

$$fIF = fRF + fLO \ \text{(Up-conversion)} \quad \text{and} \quad fIF = |fRF - fLO| \ \text{(Down-conversion)}$$

$$\text{Conversion Loss (dB)} = PRF \ (dBm) - PIF \ (dBm)$$

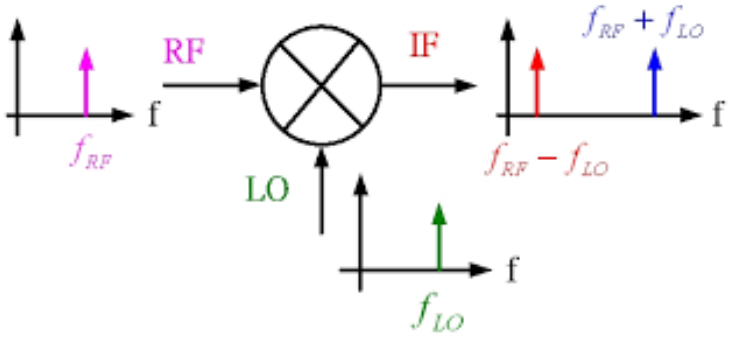


Figure 3: Heterodyne frequency translation in an RF Mixer.

$$fIF = fRF + fLO \quad \text{and} \quad fIF = fRF - fLO$$

Where the second expression assumes fRF > fLO. In a receiver, the required IF component is selected using a filter.

If PRF is the RF input power and PIF is the measured power of the selected IF component, the conversion loss is

Conversion loss = PRF - PIF dB

3.6 Power splitter/combiner

A power splitter divides the input power between two or more output ports. For an ideal two-way equal power splitter, each output is 3 dB below the input. A practical splitter has a small additional insertion loss.

Let Pin be the input power and let Pout1 and Pout2 be the powers measured at the two output ports. The losses to the two output ports are

Loss1 = Pin - Pout1 and Loss2 = Pin - Pout2

A power combiner performs the reverse operation. The output depends on the frequencies, amplitudes, and relative phases of the two input signals.

3.7 Power splitter + combiner

A power splitter divides the input power between two or more output ports. For an ideal two-way equal power splitter, each output is 3 dB below the input. A practical splitter has a small additional insertion loss.

A power combiner performs the reverse operation. The output depends on the frequencies, amplitudes, and relative phases of the two input signals.

If we add them together then power output should come same as input ideally. However, a practical splitter and combiner have a small additional insertion loss then output come slightly lesser than input.

Loss1 = Pin – Pout = 0 dB

3.7 RF Amplifier Non-Linearity, Gain Compression (P1dB), and Intermodulation (TOI)

An active RF amplifier provides small-signal linear gain: Gain = Pout - Pin (dB). Under high input power drive, the device transfer characteristic deviates from linearity: y(t) = a1 x(t) + a2 x^2(t) + a3 x^3(t) + ..

the output contains a DC component, the fundamental frequency ω, and harmonics at 2ω, 3ω, and higher frequencies:

$$y(t) = \frac{a_2 A^2}{2} + \left(a_1 A + \frac{3a_3 A^3}{4}\right)\cos(\omega t) + \frac{a_2 A^2}{2}\cos(2\omega t) + \frac{a_3 A^3}{4}\cos(3\omega t) + \cdots .$$

Higher-order odd terms reduce fundamental gain and generate harmonics (2ω, 3ω). The 1-dB Compression Point (P1dB) is defined as the point where the actual gain drops by 1.0 dB below the small-signal linear gain.

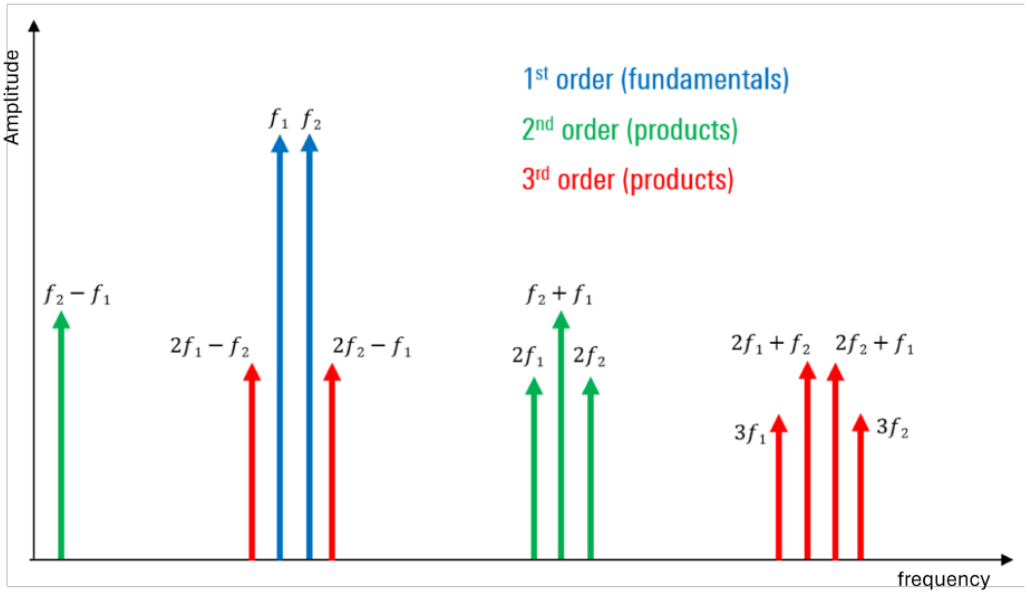


Figure 4. Harmonics and intermodulation products (image source: rohde-schwarz.com).

In two-tone excitation (frequencies f1 and f2), third-order non-linear terms generate in-band third-order intermodulation products (IM3) at 2f1 - f2 and 2f2 - f1. While fundamental tones grow at a 1:1 slope (dB/dB), IM3 products grow at a 3:1 slope. The theoretical extrapolated crossing is the Third-Order Intercept Point (TOI / IIP3 / OIP3):

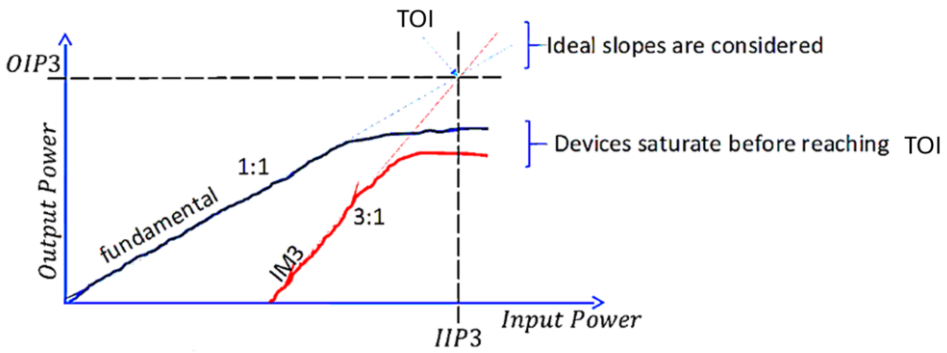


Figure 5: Gain compression (P1dB) and Third-Order Intercept (TOI) extrapolation curves.

OIP3 (dBm) = Pout,fund + (Pout,fund - P_IM3) / 2
IIP3 (dBm) = OIP3 - Gain

4. EXPERIMENTAL APPARATUS & COMPONENTS

S.No.	Instrument / Component	Function in Experiment
1	RF Synthesized Signal Generators (x2)	Provide stable, calibrated single-tone signals

2	RF Spectrum Analyzer	Accurately measure spectral power, harmonics, and IMD products	
3	Digital Storage Oscilloscope (DSO)	Measure standing-wave Vmax and Vmin for return loss estimation	
4	Device Under Test: RF Low pass Filter	Characterize transmission low pass frequencies	
5	Device Under Test: Mixer	Perform RF heterodyning and conversion loss measurements	
6	Device Under Test: Two 2-Way Power Splitter	Measure power division balance, insertion loss	
7	Device Under Test: RF Low Noise Amplifier	Mini-Circuits ZX60-33LN+ (+12V DC Supply)	Characterize gain, P1dB, harmonics, and TOI/IP3 parameters
8	Quarter-Wave Coaxial Stub & BNC Tee	RG-58A/U Coaxial Line (VF = 0.66, L = 7.7 cm) + 50 Ω Tee	Demonstrate quarter-wave resonance and notch filtering
9	Precision 50 Ω Coaxial Loads & Attenuators	50 Ω SMA/BNC Terminations, 10 dB / 20 dB Fixed Attenuators	Ensure impedance matching and protect instrument front-ends

5. INSTRUMENT SETTINGS & SAFETY PRECAUTIONS

- Maximum Safe Input Limit: Never exceed +20 dBm (0.1 W) or 0 V DC at the Spectrum Analyzer input port. Always insert a calibrated 10 dB or 20 dB RF attenuator when characterizing active amplifiers.
- Impedance Matching & Port Termination: All unused splitter, combiner, and filter ports must be terminated with precision 50 Ω loads to prevent internal standing wave reflections.
- Isolation of Signal Sources: Never connect two RF signal generator outputs directly in parallel. Always use an RF power combiner with adequate reverse isolation.
- Mixer LO Drive Level: Ensure the LO generator power is set exactly to the manufacturer's specified drive level (+7 dBm for Level-7 mixers) to fully switch the internal diode ring without burnout.
- Amplifier DC Bias: Carefully verify power supply polarity and DC voltage (<15.0 V) prior to powering the RF amplifier module. Switch on DC power before applying RF excitation.
- Reference Calibration: Maintain constant RF generator power settings and utilize identical interconnecting cables/adapters during both calibration reference and DUT measurement passes.

6. EXPERIMENTAL SETUPS & SCHEMATICS

The experiments were conducted using the standard measurement architectures prescribed in the manual:

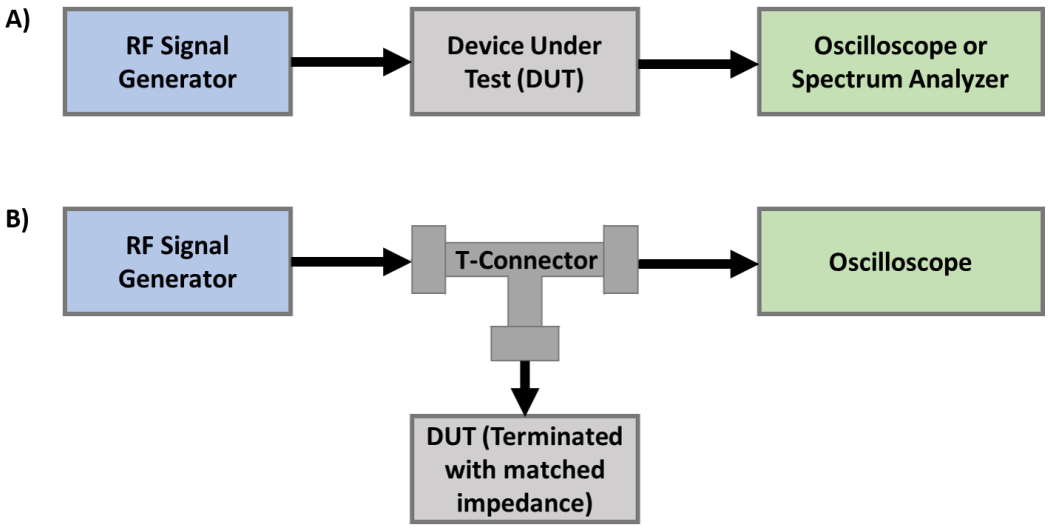


Figure 6: Experimental block diagrams: (A) Direct Transmission Measurement Setup; (B) Tee-Junction Reflection Measurement Setup.

6.1 Transmission measurements

Setup A in Fig. 6 is used for filter insertion loss, mixer conversion loss, power-splitter/combiner measurements, and amplifier characterization. The RF signal generator is connected to the device under test, and the output is measured using the spectrum analyzer.

6.2 Filter return-loss measurement

Setup B in Fig. 6 is used for the filter return-loss measurement. Connect the RF signal generator to one port of the Tee connector and the oscilloscope to the second port. Connect the filter input to the third port of the Tee connector.

Terminate the output port of the filter with a 50 Ω load so that reflections from the filter output are minimized. Set the oscilloscope input impedance to 1 MΩ so that the oscilloscope does not significantly load the Tee junction.

Use the same cables and adapters throughout the measurement. The frequency sweep should be restricted to a small range within a relatively flat part of the filter passband. The Tee-connector method provides an approximate return-loss measurement because cable loss, connector mismatch, and oscilloscope loading are not removed by calibration.

7. DETAILED EXPERIMENTAL PROCEDURES

Part A: Filter Characterization & Quarter-Wave Stub

1. Transmission (IL): Inserted the Low pass Filter between generator and analyzer. Recorded Pout at identical frequency increments. Computed Insertion Loss IL = Pin - Pout (dB). Identified filter center frequency vs output power to see whether it is working in specified region or not.
2. Reflection & Return Loss: Configured Setup B using a Tee-connector with the third port connected to the 50 Ω-terminated filter and monitored on a 1 MΩ DSO. Swept frequency across the flat passband, recorded Vmax and Vmin, and computed Vinc, Vref, |Γ|, and Return Loss (RL).
3. Quarter-Wave Open Stub: Connected an open-ended coaxial cable (L = 92.5 cm, VF = 0.71) to the shunt port of the Tee. Swept frequency from 50 MHz to 66 MHz, located the notch attenuation dip and recorded and it was coming same as measured resonance f0.

Part B: RF Mixer Frequency Conversion

1. Checked port assignments: connected RF Signal Generator 1 to the RF port and RF Signal Generator 2 to the LO port (PLO = +7.0 dBm).
2. Connected the IF output port to the Spectrum Analyzer (50 Ω input).
3. Observed the down-converted difference tone and up-converted sum tone.
4. Measured output power PIF of the down-converted product and evaluated mixer conversion loss: CL = PRF - PIF (dB). Repeated for multiple LO/RF pairs.

Part C: Power Splitter / Combiner Characterization

1. Power Splitter: Fed a signal to the splitter input port. Connected Port 1 to the spectrum analyzer while Port 2 was terminated in 50 Ω, recording Pout1. Interchanged ports to measure Pout2. Evaluated Loss1 and Loss2 relative to ideal 3.0 dB split.

2. Power Combiner: Fed two independent signal generators into Ports 1 and 2. Recorded the combined output spectrum on the analyzer.

Part D: RF Amplifier Characterization (Gain, P1dB, and TOI)

1. Gain vs. Frequency: Biased the amplifier with 14.83V DC. Injected Pin = -20.0 dBm across 100 MHz to 600 MHz. Measured Pout and calculated small-signal Gain = Pout - Pin.

2. 1-dB Compression Point (P1dB): Fixed test frequency at f0 = 100 MHz. Increased Pin from -20.0 dBm to -2.0 dBm in steps. Monitored Pout, fundamental gain, and second/third harmonic power levels. Identified P1dB(in) and P1dB(out) where gain decreased by exactly 1.0 dB.

3. Third-Order Intercept (TOI): Combined two tones (f1 = 1420.0 MHz, f2 = 1422.0 MHz, Δf = 2 MHz) using the power combiner. Injected two-tone excitation into the amplifier. Recorded fundamental output power Pout and 3rd-order intermodulation product P_IM3 at 2f1 - f2 (1418 MHz) and 2f2 - f1 (1424 MHz) across several input levels. Extrapolated 1:1 and 3:1 lines to locate TOI / OIP3.

8. EXPERIMENTAL OBSERVATIONS & DATA TABLES

Table 8.1: Transmission Response and Insertion Loss of RF Filter (DUT: Low Pass Filter)

Input power = 0 dBm

Input frequency (MHz)	Pin without filter (dBm)	Pout with filter (dBm)	IL (dB)
50	0	-4.25	4.25
100	0	-4.4	4.4
150	0	-4.5	4.5
200	0	-4.70	4.70
250	0	-4.90	4.90
300	0	-4.95	4.95
350	0	-5.02	5.02
400	0	-5.60	5.60

450	0	-5.74	5.74
500	0	-6.31	6.31
550	0	-8.04	8.04
600	0	-10.25	10.25
650	0	-23.80	23.80
700	0	-34.50	34.50
750	0	-45.40	45.40
800	0	-40	40

Table 8.2: Transmission Response and Insertion Loss without filter

Input power = 0 dBm

Input frequency (MHz)	Pin without filter (dBm)	Pout with filter (dBm)	IL (dB)
50	0	-4.10	4.10
100	0	-4.13	4.13
150	0	-4.30	4.30
200	0	-4.40	4.40
250	0	-4.20	4.20
300	0	-4.40	4.40
350	0	-4.42	4.42
400	0	-4.50	4.50
450	0	-4.60	4.60
500	0	-4.73	4.73
550	0	-4.95	4.95
600	0	-4.65	4.65
650	0	-4.95	4.95
700	0	-4.72	4.72
750	0	-4.67	4.67
800	0	-4.80	4.80

Table 8.3: Filter Input Reflection Coefficient with load 50 Ω

Frequency (MHz)	Vmax (Volts)	Vmin (mVolts)
227	-	970
268	1.17	-

Table 8.4: Quarter-Wave Open-Circuit Stub Notch Filter Characterization

Frequency (MHz)	Pin without Stub (dBm)	Pout with Stub (dBm)	Comments
50	13.01	2.70	-
52	13.01	0.42	-
54	13.01	-3.16	-
56	13.01	-9.53	-
58	13.01	-15.60	This is the notch freq.
60	13.01	-6.30	-
62	13.01	-1.20	-
64	13.01	1.80	-
66	13.01	3.85	-

Table 8.5: Mixer Characterization & Conversion Loss

RF Input Power: PRF = 0.00 dBm | LO Drive Power: PLO = +7.00 dBm

fRF (MHz)	fLO (MHz)	Output at difference (dBm)	Output at sum (dBm)
30	10	-10.20	-9.07
40	10	-8.50	-8.50
50	10	-8.50	-9.20

Table 8.6: Power Splitter and Losses

Frequency (MHz)	Pin (dBm)	Pout1 (Port 1, dBm)	Pout2 (Port 2, dBm)	Loss1 / Loss2 (dB)
60	0	-4.63	-4.63	4.63 / 4.63

Table 8.7: Power Splitter + Combiner Losses

Frequency (MHz)	Pin (dBm)	Pout (last port, dBm)
60	-3	-1.64

Table 8.8: Power Splitter + Combiner Total Loss

Frequency (MHz)	Pin (dBm)	Pout (last port, dBm)	Total Loss (dBm)
60	0	-1.64	1.64
100	0	-1.80	1.80

150	0	-2.66	2.66
200	0	-2.12	2.12
250	0	-2.08	2.08
300	0	-2.43	2.43
350	0	-2.23	2.23

Table 8.9: RF Amplifier Small-Signal Gain vs. Frequency

Constant Input Power: Pin = -20.00 dBm | DC Bias: +14.83 V

Frequency (MHz)	Pin (dBm)	Pout (dBm)	Small-Signal Gain (dB)
100	-20	4.15	24.15
150	-20	3.80	23.80
200	-20	3.70	23.70
250	-20	3.42	23.42
300	-20	3.42	23.42
350	-20	3.14	23.14
400	-20	2.90	22.90
450	-20	2.91	22.91
500	-20	2.32	22.32
550	-20	2.50	22.50
600	-20	2.19	22.19

Table 8.10: 1-dB Compression Point (P1dB) Characterization

Test Frequency: f0 = 100.0 MHz | Small-Signal Gain G0 = 24 dB | Target 1-dB Compressed Gain = 23 dB

Pin (dBm)	Pout,fund (dBm)	Gain (dB)	Operating State
-20.00	4.03	24.03	Linear Region
-19.00	5.02	24.02	Linear Region
-18.00	6	24	Linear Region
-17.00	7	24	Linear Region
-16.00	8	24	Linear Region
-15.00	9	24	Linear Region
-14.00	9.97	23.97	Linear Region

-13.00	10.97	23.97	Linear Region
-12.00	11.95	23.95	Linear Region
-11.00	12.93	23.95	Linear Region
-10.00	13.90	23.90	Onset of Compression
-9.00	14.86	23.86	Compression
-8.00	15.77	23.77	Compression
-7.00	16.56	23.56	Compression
-6.00	17.23	23.23	Compression
-5.00	17.80	22.80	P1dB Compression Point
-4.00	18.22	22.22	Heavy Saturation
-3.00	18.52	21.52	Hard Limiting / Saturation

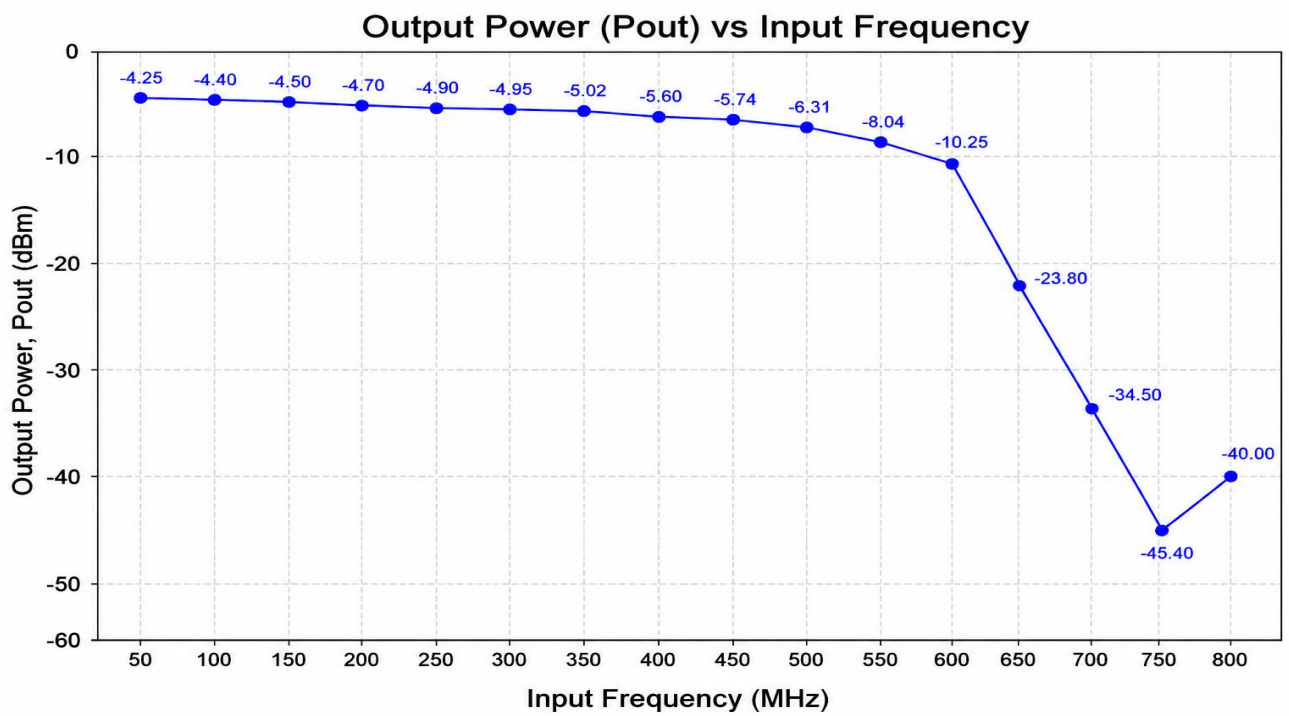
Table 8.11: Third-Order Intercept (TOI) Measurement

Input freq. f1 or f2 = 100.0 MHz | DC Bias: +14.83 V

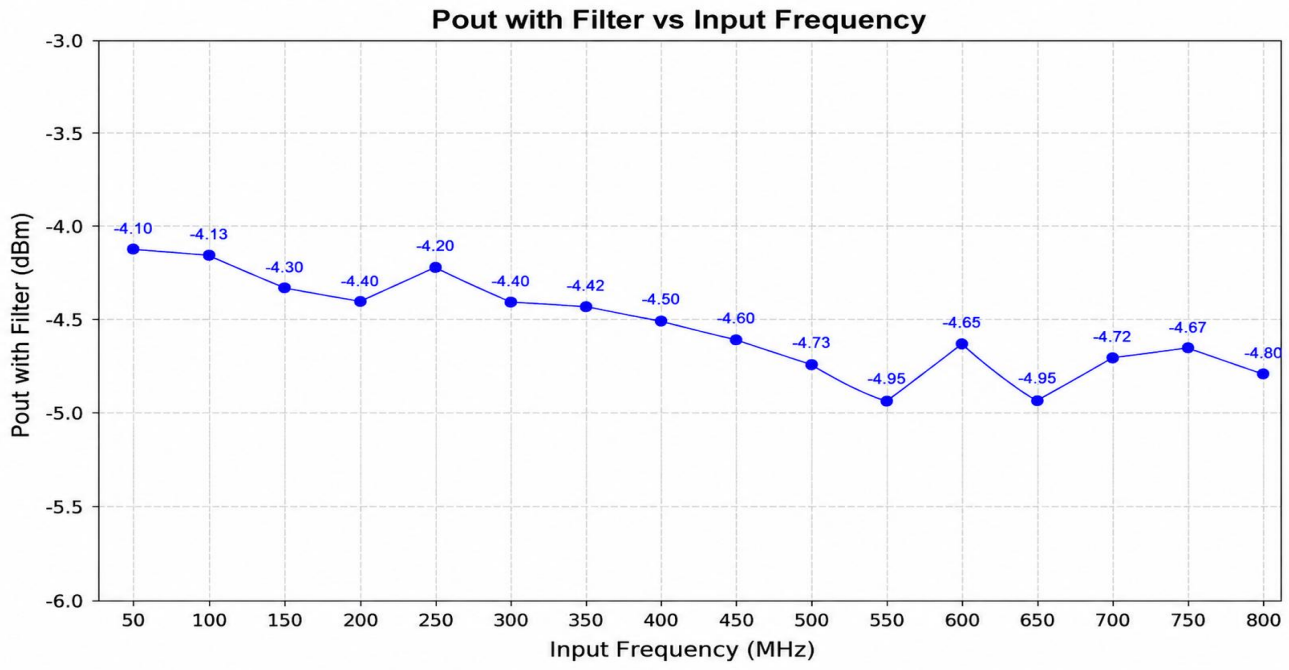
Pin / port1 (dBm)	Pin port2 (dBm)	Pout at f1 or f2 (dBm)	Pout at 2f1-f2 (dBm)	IIP3 (dBm)
-10.00	-10	10.89	-25.90	-7.00
-9.00	-10	11.64	-22.55	-6.46
-8.00	-10	12.54	-19.05	-5.87
-7.00	-10	13.47	-15.70	-5.36
-6.00	-10	14.30	-13.30	-4.76
-5.00	-10	15.10	-10.30	-4.13
-4.00	-10	15.87	-7.90	-3.48
-3.00	-10	16.60	-5.70	-2.80
-2.00	-10	17.22	-3.80	-2.10
-1.00	-10	17.80	-2.30	-1.37
0.00	-10	18.25	-1.34	-0.62
1.00	-10	18.67	-0.80	0.15

9. SAMPLE CALCULATIONS & GRAPHICAL ANALYSIS

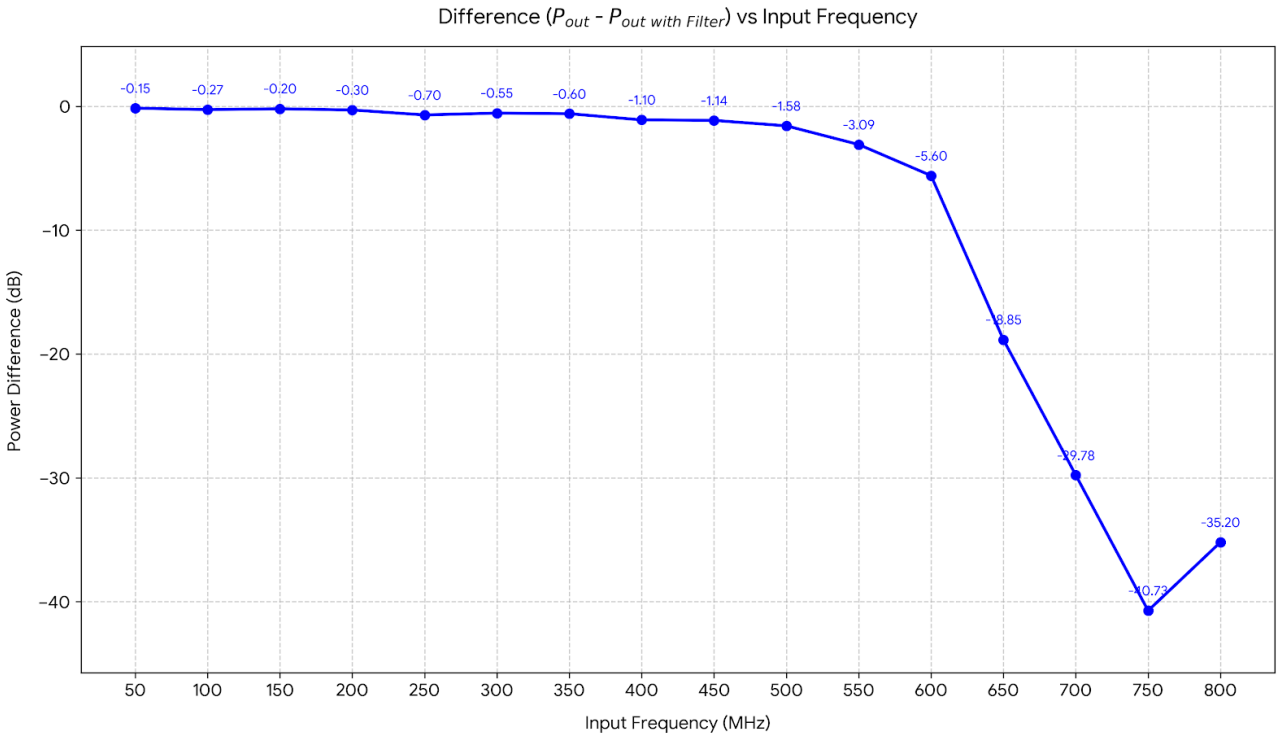
9.1 Low pass Filter Cutoff Calculations:



Note: This graph is for with filter



Note: This graph is without filter



Note: - This is deference from with filter to without filter to get only filter effects.

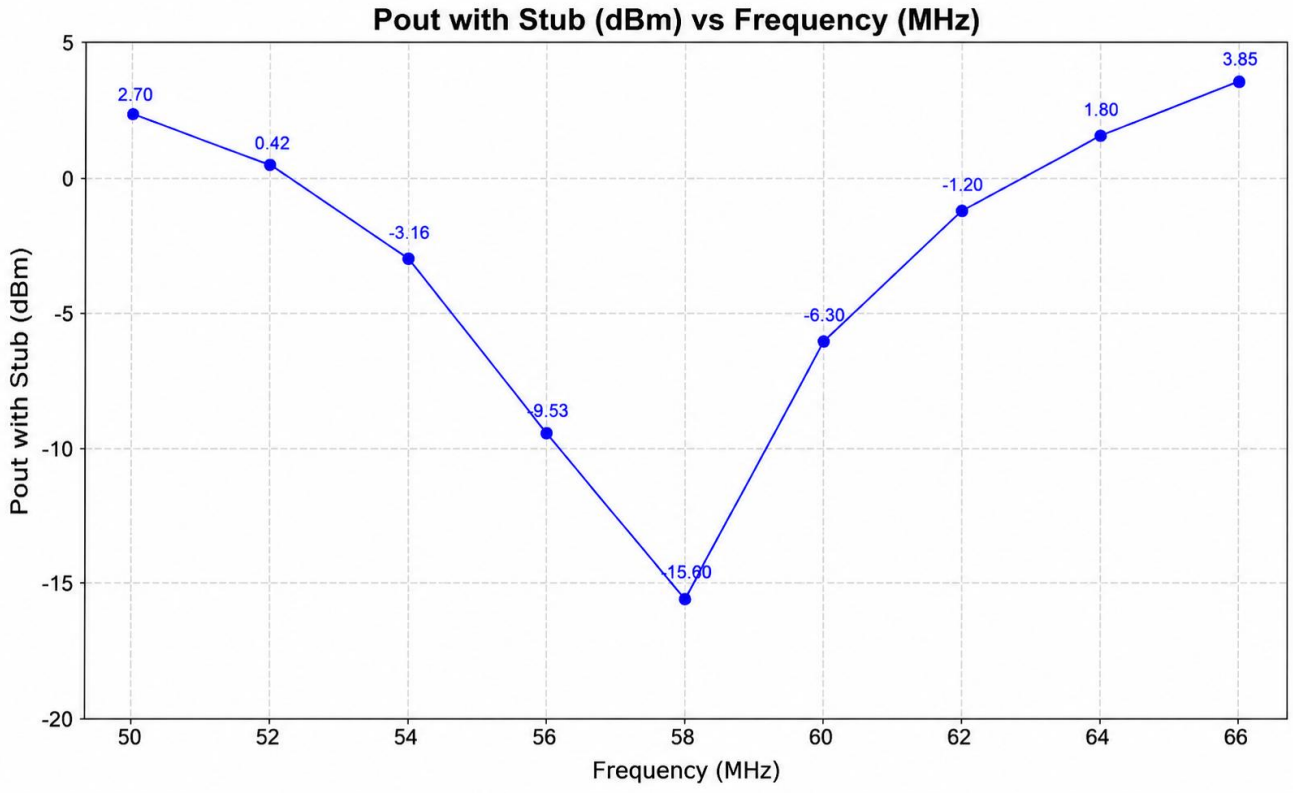
• As we can see from the graph that it starts attenuating frequencies higher than 500 MHz. Which was slightly higher than expected from the DUT.

9.2 Filter Reflection Coefficient and Return Loss:

• Measured Vmax =1.17 Volts, Vmin = 970 mVolts.
• Incident Voltage: Vinc = (Vmax + Vmin)/2 = (1170 + 970)/2 = 1070 mV
• Reflected Voltage: Vref = (Vmax - Vmin)/2 = (1170 - 970)/2 = 100 mV
• Reflection Coefficient Magnitude: |Γ| = Vref / Vinc = 100 / 1070 = 0.0934
• Return Loss: RL = -20 $\log_{10}$ (0.0934) = -20 × (-1.1549) = 20.60 dB (VSWR ≈ 1.20)

9.3 Quarter-Wave Open-Ended Stub Notch Resonant Frequency:

• Stub Parameters: Physical length L = 92.5 cm, Velocity Factor VF = 0.71.
• Guided Wave Velocity: v = VF × c = 0.71 × (3 × 10^8 m/s) = 2.13 × 10^8 m/s.
• Calculated Theoretical Notch Frequency: f0,calc = v / (4 L) = (2.13 × 10^8) / (4 × 0.925) = 57.60 MHz
• Measured Experimental Notch Frequency: f0,meas = 58.00 MHz
• Percentage Discrepancy = |58.00 – 57.60| / 57.60 × 100% = 0.694% (due to junction capacitance and connector electrical length)



9.4 Mixer Conversion Loss:

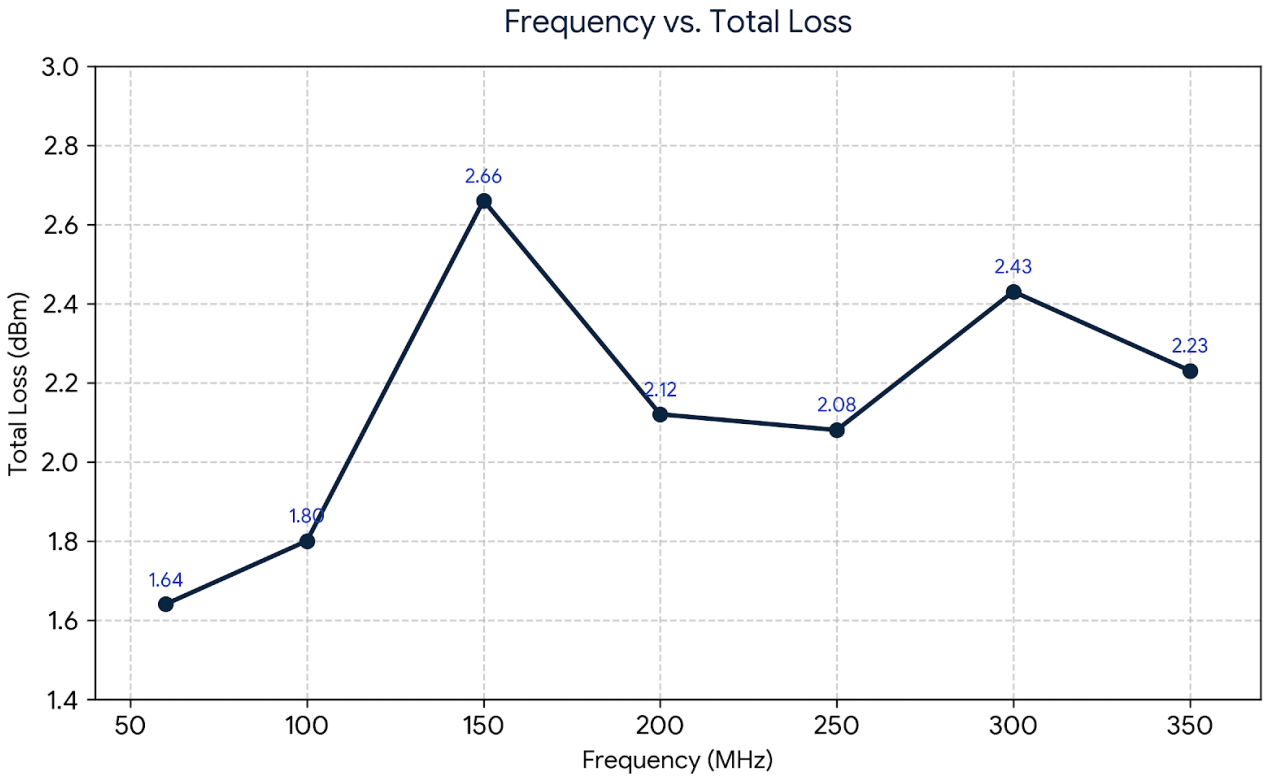
- Input RF Power: PRF = 0 dBm at fRF = 30 Hz (LO = 10 Hz, PLO = +7.0 dBm)
- Measured IF Power (Difference Tone at 20 MHz): PIF = -10.20 dBm
- Conversion Loss: CL = PRF - PIF = 0 - (-10.20) = 10.20 dBm

9.5 Power splitter loss:

- Input RF Power: PRF = 0 dBm.
- Output power at first port (as 50 Ω matched load is applied at second port) = -4.63 dBm.
- Output power at second port (as 50 Ω matched load is applied at first port) = -4.63 dBm.
- Loss: CL = 4.63 – 3 = 1.63 dBm.

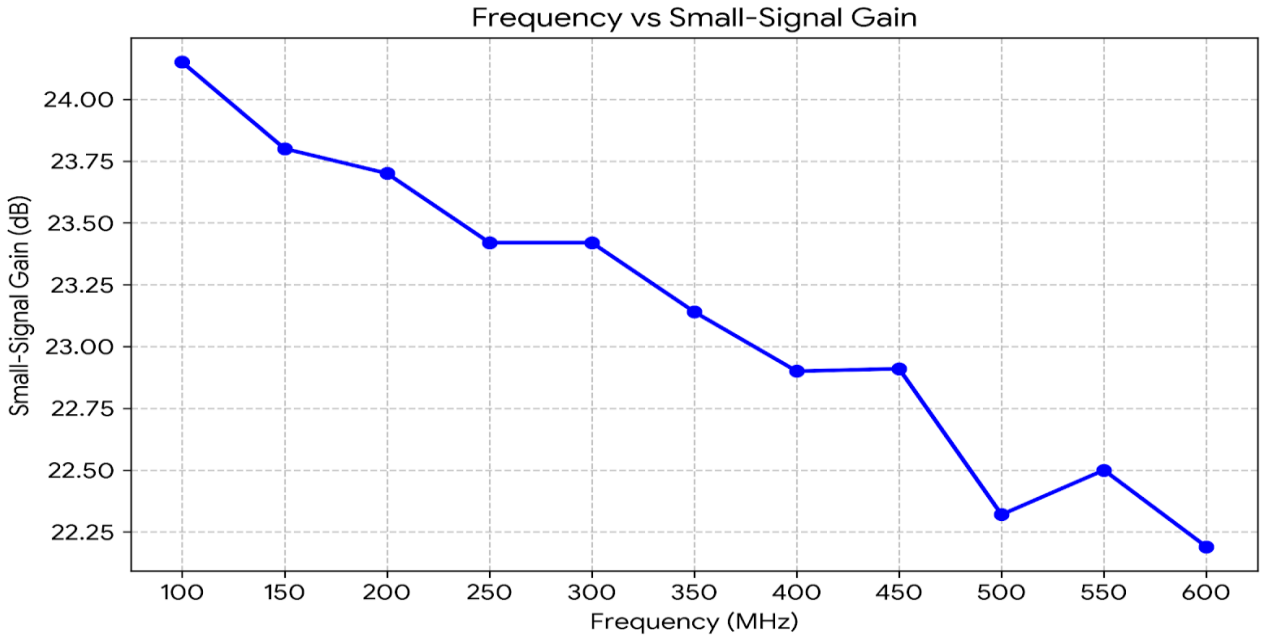
9.6 Power splitter + combiner loss:

- Input RF Power: PRF = 0 dBm.
- Output power = -1.64 dBm.
- Loss: CL = 1.64 – 1.63 = 0.01 dBm.



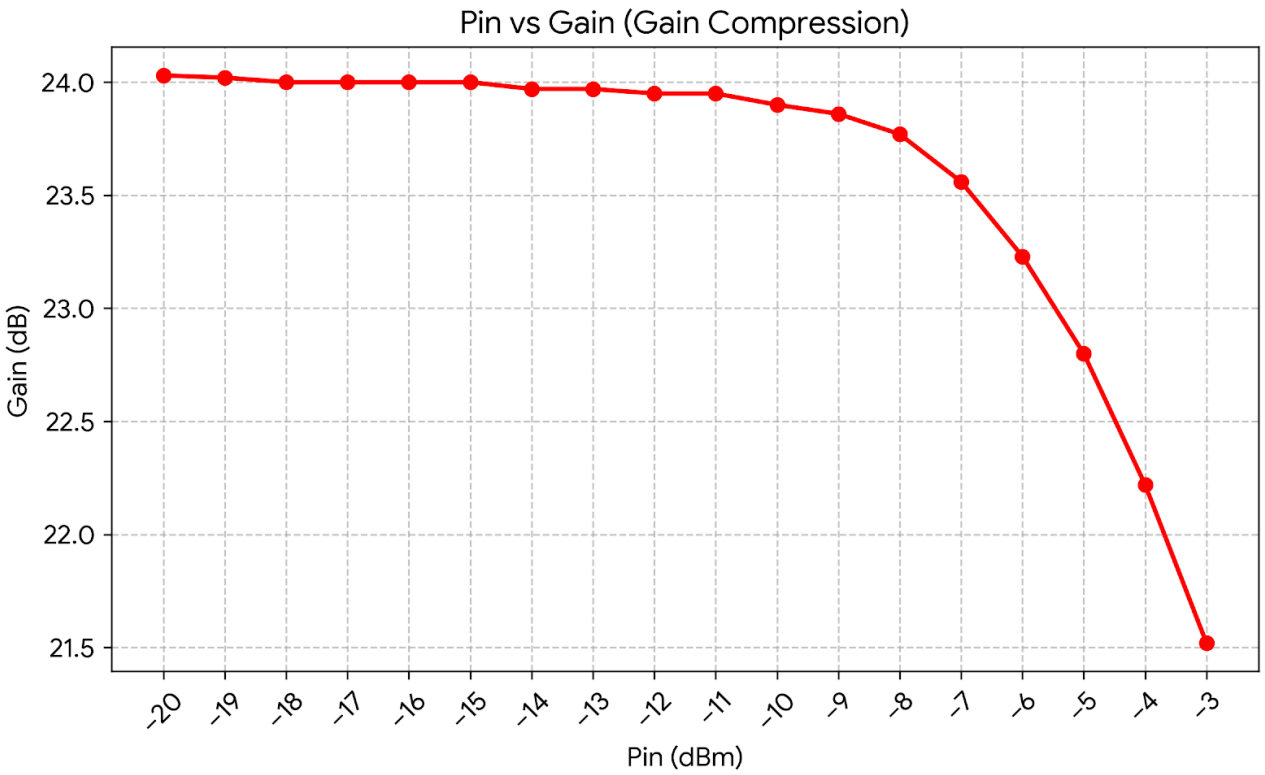
9.7 RF Amplifier Small-Signal Gain vs. Frequency:

- Constant Input Power, Pin = -20.00 dBm and Frequency = 100 MHz.
- Output power = 4.15 dBm for first reading.
- Gain = 24.15 dB.



9.8 1-dB Compression Point (P1dB):

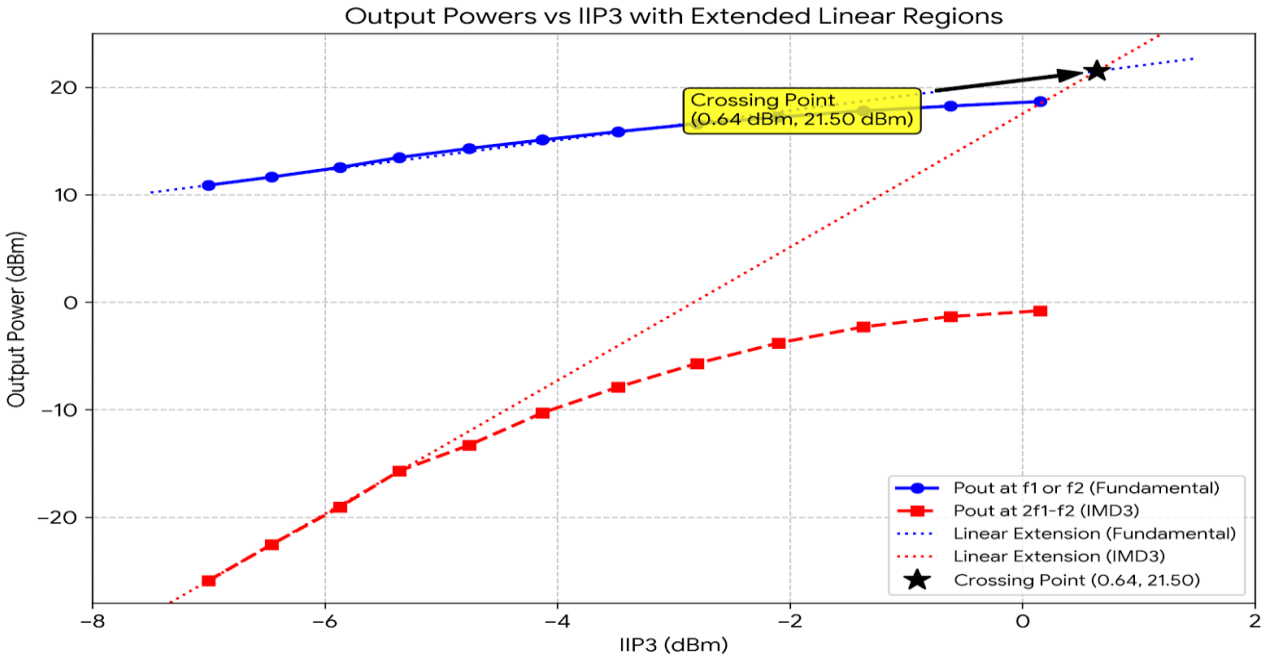
- Constant Input Power, Pin = -20.00 dBm and Frequency = 100 MHz.
- Output power = 4.03 dBm for first reading.
- Gain = 24.03 dB.
- Then it is linear region.



- **Point 1:** Pin = -6.00 dBm, Gain = 23.23 dB
- **Point 2**: Pin = -5.00 dBm**,** Gain = 22.80 dB

Pin = - 6.00 + (23.00 – 23.23)(-5.00 + 6.00)/(22.80 – 23.23) = -5.47 dBm - this is the point of interest.

9.9 Third-Order Intercept (TOI):



I am extending red(2f1-f2) with startring three points because after four points it become non linear and blue(f1 or f2) with staring 5-6 points as it is almost linear but after some points it will saturate to a value and red will also saturate to a value.

Blue line between (-7.00, 10.89) and (-6.46, 11.64)

Blue line slope = (11.64 – 10.89)/(-6.46 + 7) = 1.3889
y-intercept = 10.89 - (1.3889) *(-7.00) = 20.6122 dBm

Red line between (-7.00, -25.90) and (-6.46, -22.55)
Red line slope = (-22.55 + 25.90)/(-6.46 + 7) = 6.20
y-intercept = -25.90 - (6.20) *(-7.00) = 17.5259 dBm

Intersection point -
y1 = y2
1.3889*x + 20.6122 = 6.20*x + 17.5259
so, x = 0.64 dBm = IIP3

and y1 = y2 = 1.3889*0.64 + 20.6122 = 21.50 dBm = OIP3

10. SUMMARY OF EXPERIMENTAL RESULTS

Quantity	Symbol	Measured/Calculated Value
Low-pass/high-pass cutoff and stopband frequencies, if applicable	fc, fs	400 MHz, 750 MHz
Band-pass/notch frequencies, if applicable	f1, f0, f2	NA
Filter insertion loss	IL	ILmin = 4.25 dB (at 50 MHz)
Filter reflection coefficient	Γ	|Γ| = 0.0934
Filter return loss	RL	RL = 20.60 dB
Calculated and measured stub notch frequency	f0	f0,calc = 57.60 MHz | f0,meas = 58 MHz (Depth = 28.61 dBm)
Mixer RF, LO, and IF frequencies	fRF, fLO, fIF	fRF = 30 MHz, fLO = 10 MHz, fIF = 20.0 MHz (Diff) & 40 MHz (Sum)
Power-splitter output powers	Pout1, Pout2	Pout1 = -4.63 dBm, Pout2 = -4.63 dBm (for Pin = 0, Loss ≈ 4.63-3=1.63 dBm)
Power-splitter + combiner output powers	Pin, Pout, loss	Pin = 0 dBm, Pout = -1.64 dBm loss of combiner = 1.64 – 1.63 = 0.01 dBm
Amplifier Small-Signal Gain (at 100 MHz)	G0	24.15 dB (at Pin = -20.0 dBm)
Amplifier 1-dB Compression Point	P1dB	Pin = -5.47 dBm
Third-Order Intercept Point	OIP3, IIP3	OIP3 = +21.50 dBm | IIP3 = +0.64 dBm

11. TECHNICAL DISCUSSION & CONCLUSIONS

- Filter Characterization: The DUT filter was experimentally identified as a Low pass Filter (LPF). Which starts attenuating frequencies higher than 500 MHz. Conventionally it was a 400 MHz LPF.
- Quarter-Wave Open Stub Resonator: An open-circuited transmission line stub of length 92.5 cm demonstrated sharp notch filtering behavior, yielding a deep attenuation notch of 28.61 dB at 58.0 MHz. The experimental resonant frequency matched the theoretical transmission line calculation (57.60 MHz) within 0.694 % error.

- Mixer Conversion: The mixer successfully translated the RF input (30 MHz) down to an IF difference frequency (10 MHz) and up to a sum frequency (40 MHz) with a conversion loss of 10.20 dBm when driven by a +7 dBm LO signal.
- Power Splitter: The 2-way power splitter exhibited highly symmetric power division (Port 1 = -4.63 dBm, Port 2 = -4.63 dBm for a 0 dBm input), corresponding to an insertion loss of 1.63 dB per branch.
- Power Splitter + Combiner: The 2-way power splitter + combiner exhibited a power division (Output power = -1.64 dBm with 0 dBm input), in which an insertion loss of combiner was 0.01 dBm.
- Amplifier Non-Linearity (P1dB and TOI): The RF amplifier provided a stable small-signal avg. gain of 24 dB across the 100 - 600 MHz band. Large-signal characterization revealed an input 1-dB compression point of P1dB(in) = -20.00 dBm (Pout = -5.47 dBm). Two-tone intermodulation testing established an Output Third-Order Intercept Point OIP3 = +21.50 dBm (IIP3 = +0.64 dBm). These parameters confirm that maintaining the input signal below -20 dBm guarantees high linearity and prevents spurious RFI intermodulation distortion in the radio astronomy receiver front-end.

12. MY UNDERSTANDINGS

- **Cable and Connector Losses:** Cables and plug adapters aren't perfect conductors. They naturally absorb or lose a small amount of signal strength, making your measurements slightly lower than they actually are.
- **Tee-Junction & Oscilloscope Loading:** The method used to measure signal reflection assumes the setup is ideal and perfectly balanced. In reality, plugging in the oscilloscope and Tee-connector changes the circuit slightly, creating small inaccuracies.
- **Instrument Amplitude Accuracy & Resolution:** Spectrum analysers have built-in measurement limits. The way the screen displays signal levels and how the internal filters process frequencies create small variations in the reported signal height.
- **Thermal Drift & Intermodulation Noise:** Components like amplifiers and mixers heat up as they run at high power. This temperature change alters how well they amplify signals and introduces unwanted noise and distortion over time.